

Translating Sine and Cosine Graphs

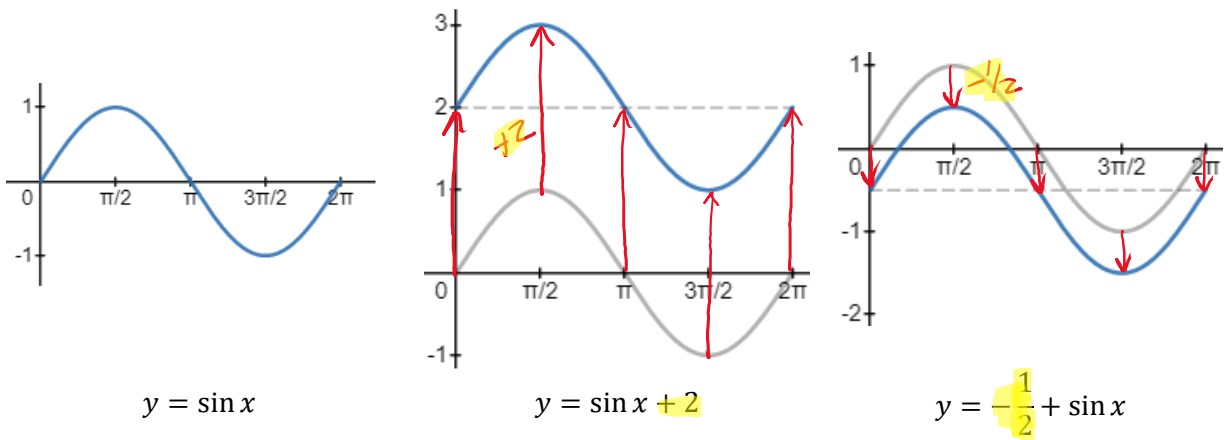
Like any function, the graph of sine and cosine, as well as the other four trigonometric functions, can be translated up, down, left, and right using addition and subtraction.

Vertical Translations

IMPORTANT INFO: Vertical Translations

Given the value $d > 0$, and any function $f(x)$, the graph of $y = f(x) + d$ is a translation d units upward from the graph of $y = f(x)$. Likewise, the graph of $y = f(x) - d$ is a translation d units downward.

Here we see some examples of vertical translation for $y = \sin x$.

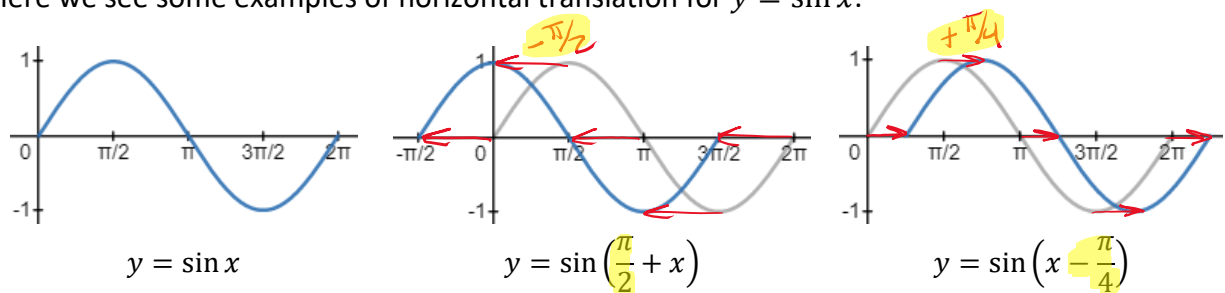


Horizontal Translations

IMPORTANT INFO: Horizontal Translations

Given the value $c > 0$, and any function $f(x)$, the graph of $y = f(x + c)$ is a translation c units left from the graph of $y = f(x)$. Likewise, the graph of $y = f(x - c)$ is a translation c units right.

Here we see some examples of horizontal translation for $y = \sin x$.



EXAMPLE 1

Identify any horizontal and/or vertical translation for the functions below.

(a) $y = 3 \sin(x - \pi) - 1$

Right π units

Down 1 unit

(b) $y = \frac{1}{2} - 2 \cos\left(\frac{\pi}{2} + x\right)$

Left $\frac{\pi}{2}$ units

Up $\frac{1}{2}$ unit

Combining Transformations**IMPORTANT INFO: Summary of Transformations**

Consider $y = a \sin[b(x + c)] + d$ or $y = a \cos[b(x + c)] + d$, with $b > 0$.

Amplitude:	The value $ a $ is the amplitude. If $a < 0$, the graph will be reflected across the sinusoidal axis.
Period:	The value $2\pi/b$ is the period.
Horizontal Shift:	If $c > 0$, the graph is shifted left c units. If $c < 0$, the graph is shifted right $ c $ units. (Note some texts refer to this as phase shift.)
Vertical Shift:	If $d > 0$, the graph is shifted up d units. If $d < 0$, the graph is shifted down $ d $ units.

EXAMPLE 2

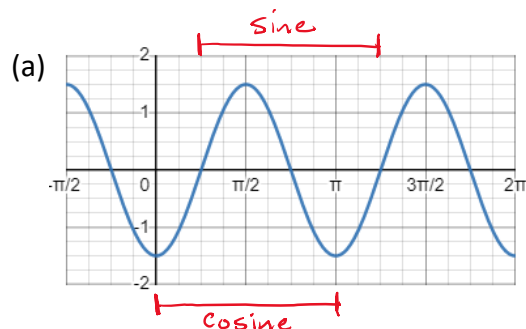
Complete the table.

Function	Parent Function	Period	Amplitude	H. Shift	V. Shift
$y = 2 + \cos\left(\frac{1}{2}\theta\right)$	$y = \cos \theta$	$\frac{2\pi}{1/2} = 4\pi$	1	n/a	Up 2
$y = -1 - \sin\left(\theta - \frac{\pi}{7}\right)$	$y = \sin \theta$	2π	1 * Reflect over H. axis	Right $\pi/7$	Down 1
$y = \frac{1}{3} \sin(2\theta + \pi) + 4$	$y = \sin \theta$	$\frac{2\pi}{2} = \pi$	$\frac{1}{3}$	Left $\pi/2$	Up 4

$y = \frac{1}{3} \sin\left(2\left(\theta + \frac{\pi}{2}\right)\right) + 4$

EXAMPLE 3

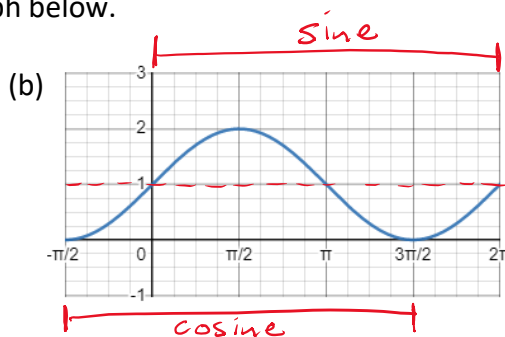
Write both a sine and cosine rule for each graph below.



Period = π
Amplitude = 1.5

$$y = -1.5 \cos(2x)$$

$$y = 1.5 \sin\left(2\left(x - \frac{\pi}{4}\right)\right)$$



Period = 2π
Amplitude = 1

V. shift up 1

$$y = \sin(x) + 1$$

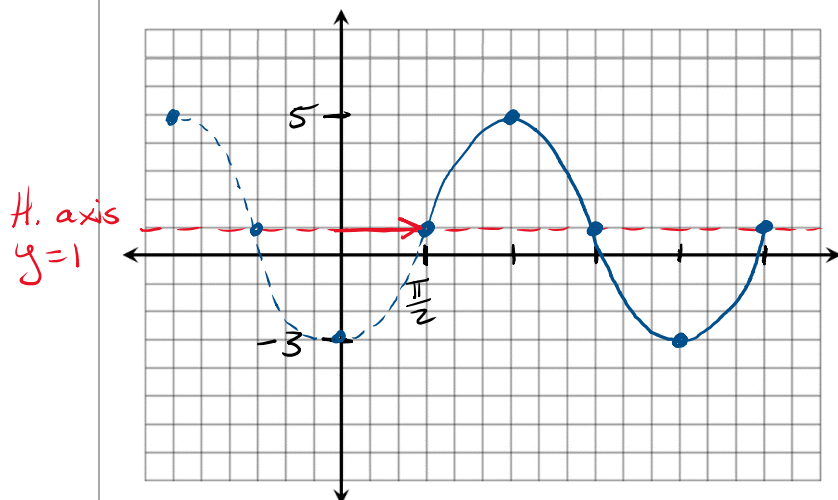
$$y = -\cos\left(x + \frac{\pi}{2}\right) + 1$$

H. axis
 $y = 1$

Answers
vary...

EXAMPLE 4

Determine the amplitude, period, and any other transformations for the function $f(x) = 4 \sin(x - \pi/2) + 1$, then graph one period.



Amplitude = 4

Period = 2π

Step = $\frac{2\pi}{4} = \frac{\pi}{2}$

H. shift right $\frac{\pi}{2}$

V. shift up 1 → H. axis
 $y = 1$

Complications Involving Horizontal Translation

Graphing a trigonometric function that includes a horizontal translation can be challenging, and choosing an appropriate horizontal scale can sometimes be problematic. You should consider several factors in making this decision: the period, horizontal spacing between points of interest (e.g., maximums, minimums, and intercepts), and the amount of horizontal translation. One good practice is to express each of these quantities in a way that utilizes a **common denominator**.

EXAMPLE 5

Identify all transformations and construct a graph.

$$f(x) = \sin(3x - \pi)$$

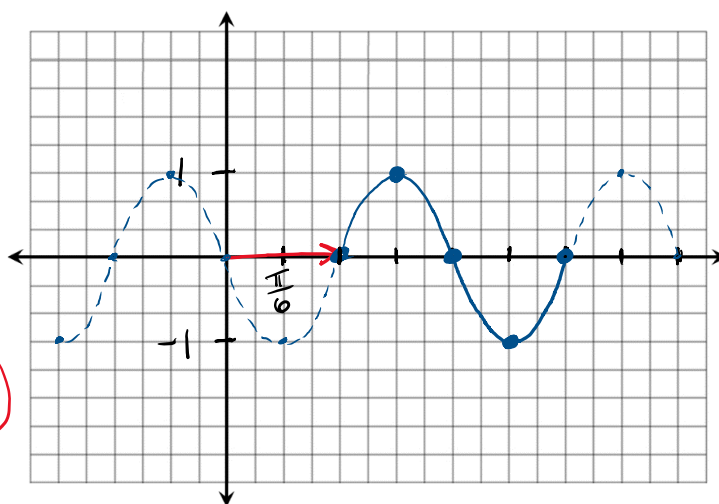
$$f(x) = \sin\left(3\left(x - \frac{\pi}{3}\right)\right)$$

$$\text{Amplitude} = 1$$

$$\text{Period} = \frac{2\pi}{3}$$

$$\text{Step} = \frac{2\pi/3}{4} = \frac{2\pi}{12} = \frac{\pi}{6}$$

$$\text{H. shift right } \frac{\pi}{3} = \frac{2\pi}{6} = 2\left(\frac{\pi}{6}\right)$$



$$f(x) = 2 \cos\left(\frac{\pi}{3} + \frac{x}{2}\right)$$

$$f(x) = 2 \cos\left(\frac{1}{2}\left(x + \frac{2\pi}{3}\right)\right)$$

$$\text{Amplitude} = 2$$

$$\text{Period} = \frac{2\pi}{1/2} = 4\pi$$

$$\text{Step} = \frac{4\pi}{4} = \pi = \frac{3\pi}{3} = 3\left(\frac{\pi}{3}\right)$$

$$\text{H. shift left } \frac{2\pi}{3} = 2\left(\frac{\pi}{3}\right)$$

